



One of the precautions you should take while using a resistor is not to bend its terminals too close to its body

for basic electronic purposes, resistors having power rating of $\frac{1}{4}$ W would suffice.

To measure the value of resistance of a given resistor, you could use two methods. First one is to read the colour code given on the value of resistance. The other method, and also the easier and faster one is to use a multimeter.

2.2.2 Potentiometer

The resistors we have discussed above have constant resistances, which do not vary unless they are heated too much. A potentiometer is also a resistor, but it has variable resistance which can be changed at will by the user. Potentiometer is rated according to a range of resistances which can be achieved by using it, along with its power rating. A potentiometer is very useful in applications in which you are not sure of the exact value of the resistance you would require. When only the range required is known, a potentiometer is used and then its resistance is varied so as to get desired output from the circuit.

The resistance of the potentiometer is varied by rotating the 'shaft' or 'knob' located on its body. Rotating it in one direction increases the resistance, while rotating in opposite direction decreases it.

A potentiometer has three terminals. The resistance between the extreme two terminals is constant, whereas the resistance between either of the end terminals and the middle terminal is variable.

